TIME SERIES ANALYSIS

Dr. Babji Srinivasan

Associate Professor
Department of Applied Mechanics
IIT Madras

What is a time-series?

Time-series is a set of data observed (or measured) at regularly or irregularly spaced time instants

- The independent dimension in general could be different from time, such as length, frequency or any other quantity
- Time-series is synonymous with a set of **sampled-data**
- Observations could be a result of measurements using a physical sensor (e.g., temperature, pressure, rainfall level) or by calculations of some indices (e.g., stock market price, student grades)

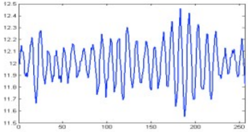
Time-series analysis

Time-series analysis consists of processing the observed data to extract meaningful and useful information. This information is used for various purposes such as forecasting (prediction), control, pattern recognition, fault detection, *etc*.

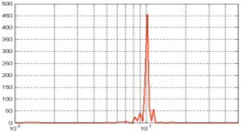
Examples of time-series

- The independent quantities (time, frequency or space) are known as <u>dimensions</u>.
- Signals can be 1-dimensional, 2-D, 3-D or more.

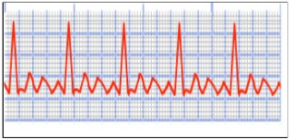
Valve output from a control loop as a function of time



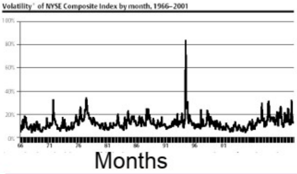
Valve output (power) as a function of frequency



ELECTROCARDIOGRAM (ECG)

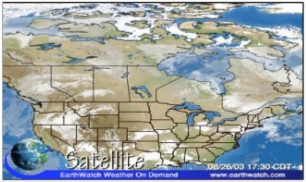


Volatility of NYSE Composite Index from 1966-2001



2-D Image of Mars on June 04, 2003

3-D Satellite Image of Canadian weather condition



Consider two sequences of observations:

(i) In a roulette game played in a casino, let $y_1, y_2, \ldots, y_{100}$ be the outcomes of 100 rounds.

(ii) Let $y_1, y_2, \ldots, y_{100}$ be the closing prices of a stock for 100 consecutive days.

Difference between the nature of outcomes of two experiments?

- Classical Statistical Inference procedures $\Rightarrow$ Independent observations are obtained under identical conditions (identically independently distributed)
- Time series data $\Rightarrow$ Chronological sequence of observations recorded at specific time intervals such as hours, months, or years etc. over a period of time
- Time series data at adjacent points are dependent

Is dependence of observations a curse or advantage?

Parallels to Statistical (Digital) Signal Processing

- Time-series is synonymous with a digital signal
- Time-series analysis in the modern times is considered equivalent to **Statistical Digital Signal Processing** (DSP)
  - Some differences exist at certain levels but nevertheless for all practical purposes, the techniques involved are identical
- Students with engineering background can treat this as DSP, while students from management, finance and non-engineering background can treat this as time-series or data analysis
- In this course, we will use the pair of phrases 'signal' and 'time-series', and 'time-series analysis' and 'data analysis' interchangeably

What is Time Series?

A Time series is a set of observations, each one being recorded at a specific time. (Annual GDP of a country, Sales figure, etc)

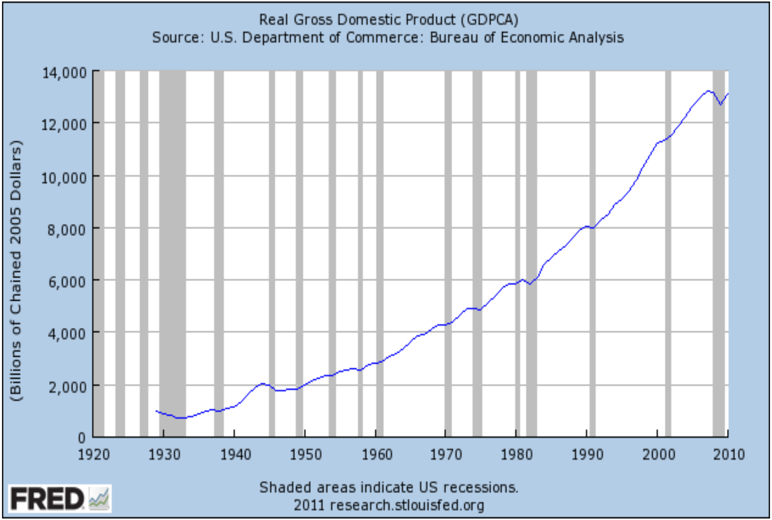
A discrete time series is one in which the set of time points at which observations are made is a discrete set. (All above including irregularly spaced data)

Continuous time series are obtained when observations are made continuously over some time intervals. *It is a theoretical Concept*. (Roughly, ECG graph).

A discrete valued time series is one which takes discrete values. (No of accidents, No of transaction etc.).

Few Time series Plots

Annual GDP of USA



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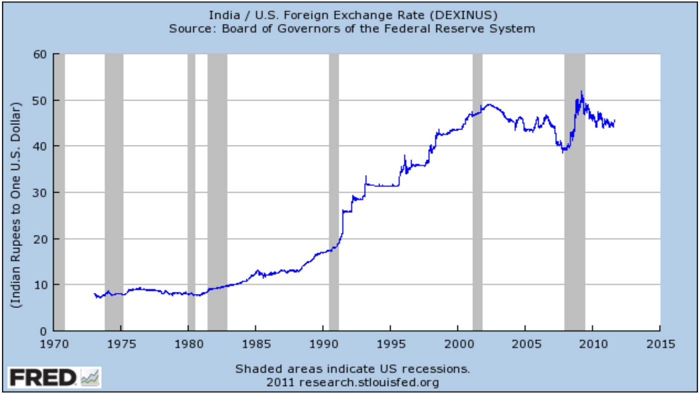
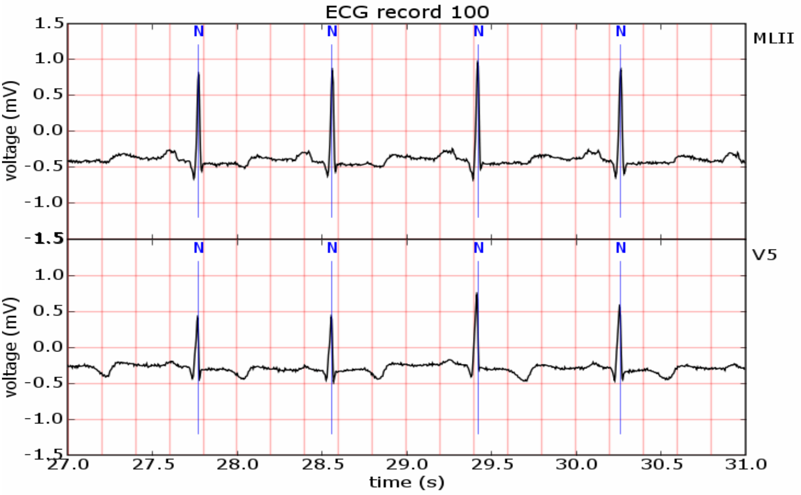


Continuous time series are obtained when observations are made continuously over some time intervals. (ECG graph).



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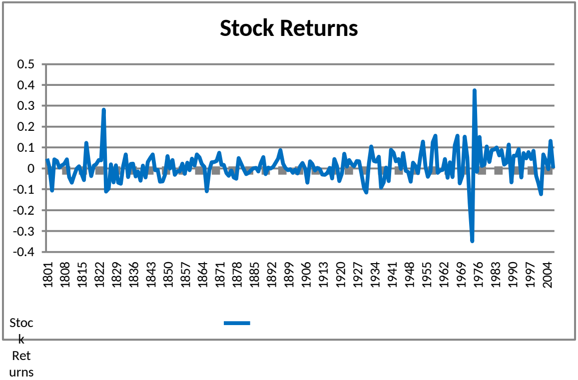
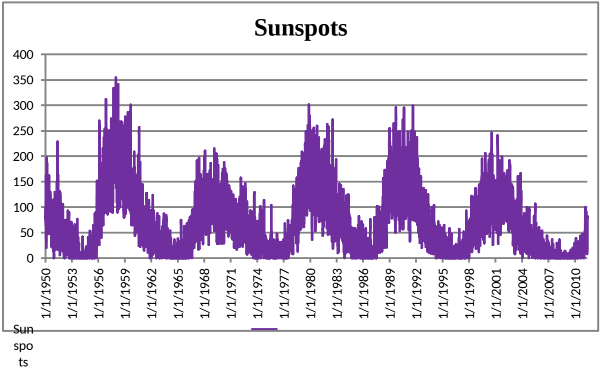
Time series plot on car accident in U.K.

Continuous time series data (Stock returns):

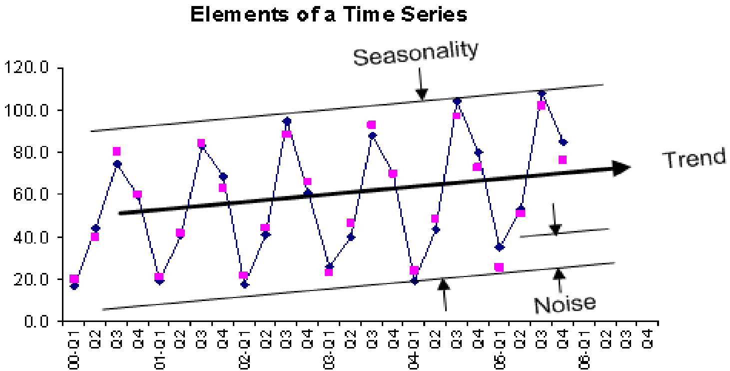


Time series data (Number of sunspots) showing cycles:



Quarterly Sales of Ice-cream

Q1-Dec-Jan



Objective of Time Series Analysis

- Forecasting (Knowing future is our innate wish).
- Control (whether anything is going wrong, think of ECG, production process etc)
- Understanding feature of the data including seasonality, cycle, trend and its nature. Degree of seasonality in agricultural price may indicate degree of development. Trend and cycle may mislead each other (Global temperature may be an interesting case)

What does time-series analysis involve?

The analysis of time-series data involves one or more of the following tasks

- **Trend analysis:** Does the observed data exhibit any particular trend such as periodicity, seasonal variation, or any other pattern?
- **Correlation analysis:** Is there a dependence between successive observations of a quantity? Is there a relationship between the evolution of one time-series and that of another time-series?
- **Spectral analysis:** What frequency components make up the observed data? What are their relative strengths? Do these frequency components vary with time?

Note that while correlation analysis primarily involves processing in time-domain, spectral analysis involves processing in frequency-domain. These are two different domains that provide complementary information.

Tasks in time-series analysis

Time-series modelling: Build models to predict the behaviour of random quantities. These models assume that a class of random signals can be modelled as fictitious random inputs passing through a linear time-invariant system.

Estimation: What is the true signal underneath a measured quantity that is in general corrupted with noise? Consists of one or more of **smoothing**, **filtering** and **prediction**. Estimation is a very important task in data analysis.

Data pre-processing: Measured data can come with outliers and many a time with missing observations. This step is concerned with detecting these outliers and suitably replacing the missing data. This step typically consumes

Important aspects of time-series modelling

- Time-series analysis is traditionally concerned with analysis of what are known as random signals
- Random signals are assumed to be generated by random
- processes Random process is essentially a process (or quantity) that evolves in time according to some probabilistic laws
- Time-series modelling is essentially concerned with prediction of these random signals from their observations
- The uniqueness of time-series modelling is that no physical input(s) can be identified as responsible for the variation in the random signals. Rather, these variations are assumed to be due to the effect of some fictitious (random) inputs.
- In contrast, in modelling of engineering systems, there is always a set of input-output pair whose relationship is sought (system identification)

Scope of applications

- Geosciences and meteorology (weather forecasting, trends in weather patterns)
- Business and finance (econometrics) (stock market analysis, business forecasting)
- Process control and other engineering applications (modelling of disturbance effects, uncertainties)
- Biostatistics
- Multivariate statistical data analysis (fault detection, pattern recognition, image analysis)
- Medicine (monitoring of lab variables, epidemic analysis, clinical decisions)
- Physics, astronomy and several other areas

Focus of this course

This course primarily focuses on theory and application aspects of the four main branches of time-series analysis, namely

1. Correlation analysis: Auto-correlation function (ACF), partial auto-correlation function (PACF), cross-correlation function (CCF)
2. Spectral analysis: Fourier transforms, power spectrum, estimation of spectrum for random and deterministic signals
3. Time-series modelling: Introduction to random signals, random processes, auto-regressive moving-average (ARMA) models, estimation of order and parameters of ARMA models
4. Filtering: Introduction to filtering, least squares theory, digital filters and Kalman filter

Necessary background on sampling theory, linear time-invariant (LTI) systems and z-transforms will be provided at appropriate junctures.

References

1. *G. Box, G.M. Jenkins and G. Reinsel (1994).* Time Series Analysis: Forecasting & Control. Prentice Hall India, 3/e.
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3. *R.H. Shumway and D.S. Stoffer (2000).* Time-Series Analysis and its Applications. Springer-Verlag, New York.
4. *J.G. Proakis and D.G. Manolakis (2005).* Digital Signal Processing - Principles, Algorithms and Applications, Prentice Hall Inc., New Jersey, USA
5. *Steven W. Smith (1999).* The Scientist and Engineer's Guide to Signal Processing, California Technical Publishing, San Diego, California, USA. (free download from `http://www.dspguide.org`)